Stress-Detection-Visualizer

Real-time emotional state detection using ESP32 + HRV/GSR sensors, classified via Python and visualised as a coloured body silhouette in TouchDesigner. Real-time stress detection and biofeedback visualisation using biosensors and generative media.

What it does

Places a finger on the sensor pad → ESP32 reads your pulse and sweat response → Python classifies your emotional state (Zen / Stress / Flow / Burnout) → TouchDesigner projects your silhouette in a matching colour onto a wall in real time.

Stack

- **Hardware:** ESP32, MAX30102 pulse sensor, GSR sensor
- **Firmware:** C++ (Arduino IDE)
- **Middleware:** Python (`osc.py`) — serial parsing, baseline calibration, OSC output
- **Visuals:** TouchDesigner — difference matte silhouette, Switch TOP colour routing

Emotional States

State	Condition	Colour
Zen	High HRV, Low GSR	Violet
Stress	Low HRV, High GSR	Hot Pink
Flow	High HRV, High GSR	Cyan
Burnout	Low HRV, Low GSR	Teal